

came up with the idea. The idea took off and ended up getting him a patent #8239662 titled “Network Based Operating System Across Devices” in 2012.

But the journey from idea to patent was not an easy one since the very idea of a cloud based operating system was hard to define in 2006, even for Google executives. The management team at Google didn’t quite grasp the idea and it was rejected outright. Given the fact that it was originally written using the Mozilla Firefox internet browser application, there was an in-built resistance to the idea itself. The original design of the Chrome OS was based on a stripped down version of Linux known as PuppyLinux which had numerous restrictions which made it a tedious build.

By the end of 2007, Nelson had augmented the design of his operating system to move away from the PuppyLinux and Mozilla Firefox backbone system and migrated it to the Chrome browser system with the help of the Google Chrome team. The Chrome browser hadn’t yet been commercialised yet but was in development at Google. It would only be made publicly available in 2008. But behind it all the driving force of the Chrome OS was neither its web-app based functionality nor its lightweight Google backed networking features - Chrome OS was designed by a software developer to be a super fast operating system for developers like him.

The way to make a superfast operating system was simple - remove the entire desktop operating system from the hard disk on to the RAM memory. This would simplify the file input/output bottlenecks present in almost all operating systems and free up huge performance resources for a faster interface. Since most of the operations performed on any system are File I/O intensive and aren’t critically dependent on CPU speed, this shift to RAM based operations would execute tasks almost instantly without increasing any efforts towards the optimisation of the many other applications that make up the operating system. This shift radically improved speed and performance where application restarts such as web browsers went from a 45 second time delay to a 1 second time delay and code compiling became faster by 60 percent. This wouldn’t be possible on any hard disk. Even SSDs would only come close.

This form of system restructuring could only be possible if the risk of data loss could be eliminated - after all RAM memory isn’t meant for storing and recovery data. That’s where the cloud component came in since all of Chrome OS applications and data are stored in web-apps online, the risk of data loss could be mitigated by using secure and efficient web applications. The system essentially ran on the RAM and accessed data as it was